

Deducing [O] or [R]

Oxidation	Reduction
<u>gain</u> of oxygen	<u>loss</u> of oxygen
_____ of hydrogen	_____ of hydrogen
_____ of electron	_____ of electron
_____ in oxidation number	_____ in oxidation number

Note:

- “OIL RIG” – oxidation is loss of electron, – reduction is gain of electron

Deducing oxidation number from formula

- Deduce the oxidation number of the underlined atom

<u>S</u> O ₄ ²⁻	$x + 4(-2) = -2$ $x = +6$
<u>N</u> O ₃ ⁻	
Na ₂ <u>O</u>	
<u>S</u> ₂ O ₃ ²⁻	

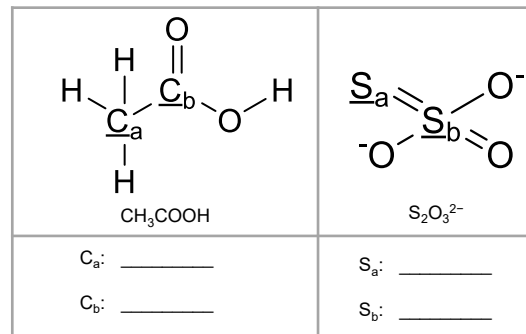
Note:

- oxidation number $\equiv$ oxidation state

Deducing oxidation number from structure

(to be covered in later topics)

- Deduce the oxidation number of the atoms in the following species



Notes:

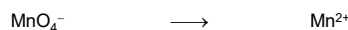
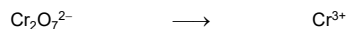
- this method allows us to deduce the specific oxidation number of each atom, instead of just the average oxidation number
- electronegativity increases from B < H < C, where C is the most electronegative

Balancing half equations

Acidic medium

- Balance the element being reduced or oxidised
- Balance O atoms by adding H₂O
- Balance H atoms by adding H⁺
- Balance the overall charges by adding e⁻

- Balance the half equation for the following reactions in acidic medium



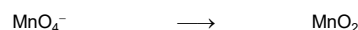
Note:

- common reduction half-equations can be found in *Data Booklet*

Basic medium

- Balance the element being reduced or oxidised
- Balance O atoms by adding H₂O
- Balance H atoms by adding H⁺
- Balance the overall charges by adding e⁻
- “Neutralise” H⁺ ions by adding OH⁻ ions to **both** sides of the half equation
- Combine the H⁺ and OH⁻ ions on same side of equation to form H₂O
- Re-balance the half equation by eliminating any common terms, such as H₂O

- Balanced the half equation for the following reaction in basic medium

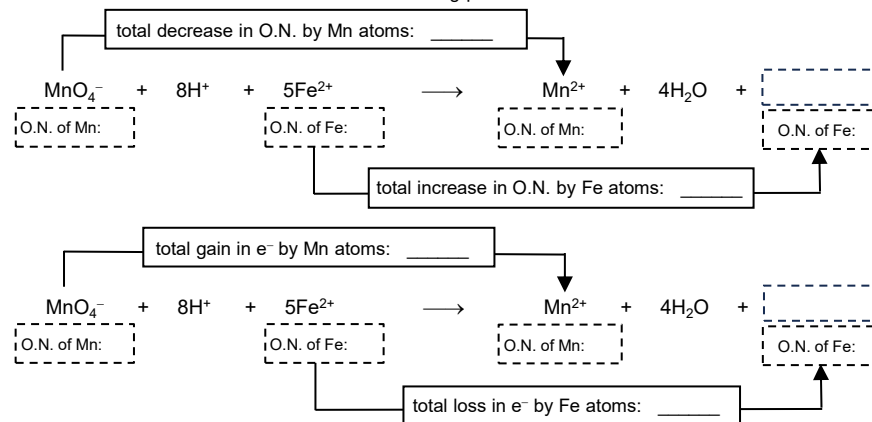


Summing up half equations

oxidation half-eq ⁿ	$2\text{I}^- \longrightarrow \text{I}_2 + 2\text{e}^-$
reduction half-eq ⁿ	$2\text{IO}_3^- + 12\text{H}^+ + 10\text{e}^- \longrightarrow \text{I}_2 + 6\text{H}_2\text{O}$
overall equation	

Solving by oxidation numbers OR e⁻

- Deduce the oxidation state of the Fe-containing product



Notes:

- when calculating total $\uparrow$ or $\downarrow$ in O.N or e⁻, must consider the number of atoms

Deducing [O] or [R]

Oxidation	Reduction
<u>gain</u> of oxygen	<u>loss</u> of oxygen
<u>loss</u> of hydrogen	<u>gain</u> of hydrogen
<u>loss</u> of electron	<u>gain</u> of electron
<u>increase</u> in oxidation number	<u>decrease</u> in oxidation number

Note:

- “OIL RIG” – oxidation is loss of electron, – reduction is gain of electron

Deducing oxidation number from formula

- Deduce the oxidation number of the underlined atom

$\underline{\text{S}}\text{O}_4^{2-}$	$x + 4(-2) = -2$ $x = +6$
$\underline{\text{N}}\text{O}_3^-$	$x + 3(-2) = -1$ $x = +5$
$\text{Na}_2\underline{\text{O}}_2$	$2(+1) + 2(x) = 0$ $x = -1$
$\underline{\text{S}}_2\text{O}_3^{2-}$	$2(x) + 3(-2) = -2$ $x = +2$

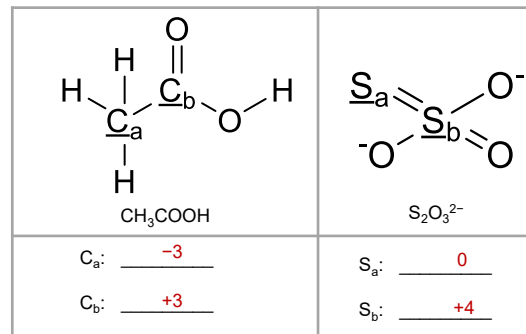
Note:

- oxidation number $\equiv$ oxidation state

Deducing oxidation number from structure

(to be covered in later topics)

- Deduce the oxidation number of the atoms in the following species



Notes:

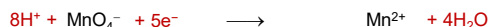
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Balancing half equations

Acidic medium

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- Balance the half equation for the following reactions in acidic medium



Note:

- common reduction half-equations can be found in *Data Booklet*

Basic medium

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- Balanced the half equation for the following reaction in basic medium

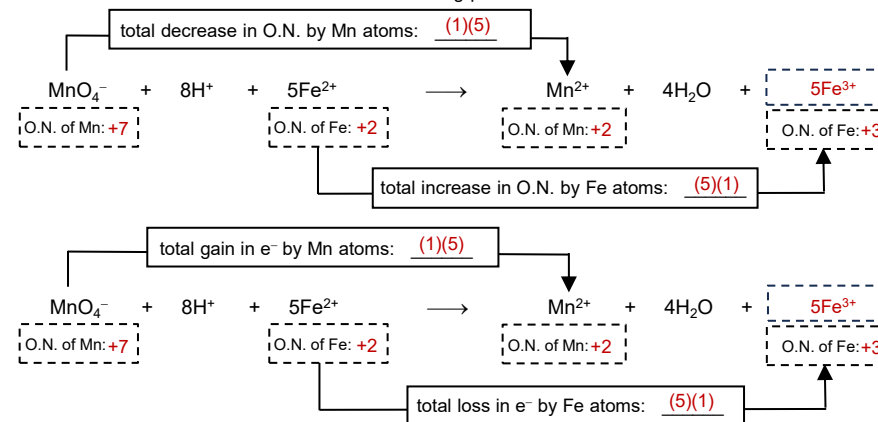


Summing up half equations

oxidation half-eq ⁿ	$2\text{I}^- \longrightarrow \text{I}_2 + 2\text{e}^-$
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overall equation	$2\text{IO}_3^- + 12\text{H}^+ + 10\text{I}^- \longrightarrow 6\text{I}_2 + 6\text{H}_2\text{O}$

Solving by oxidation numbers OR e⁻

- Deduce the oxidation state of the Fe-containing product



Notes:

- when calculating total ↑ or ↓ in O.N or e⁻, must consider the number of atoms